

Assignment: Kali Linux Essentials

Cover Page

Assignment Title: Kali Linux Essentials - Assignment 2

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Task 1: Kali Linux Fundamentals Research

1. Linux Directory Structure

Linux utilizes a hierarchical, single-tree structure starting from a root directory (represented as /). Unlike Windows, which uses drive letters (e.g., C:\, D:\) to represent separate physical or logical storage partitions, Linux mounts all drives and partitions into this single unified tree.

Important Directories:

- **/:** The root directory; the base of the entire file system.
- **/home:** Contains user-specific personal files (documents, downloads, desktop).
- **/root:** The home directory for the root (administrative) user.
- **/etc:** Stores system-wide configuration files.
- **/var:** Contains variable files, such as logs and databases, that change in size.
- **/usr:** Stores user-related programs, libraries, and documentation.
- **/tmp:** A location for temporary files created by the system or applications.
- **/opt:** Used for installing optional or add-on software packages.

here is a comparison between the Linux directory structure and the Windows directory system:

- **Structure Model:** Linux uses a single, unified hierarchical tree structure originating from a single root directory (/). In contrast, Windows uses a multi-rooted system where each storage partition, drive, or external device is assigned its own independent drive letter (such as C:\, D:\, or E:\).
- **Mounting vs. Separate Drives:** In Linux, all physical hard drives, partitions, and external media are "mounted" as subdirectories somewhere within the single root tree. In Windows, each partition or storage device remains isolated under its own drive letter root.
- **Path Separators:** Linux uses forward slashes (/) for defining file paths (e.g., /home/user/documents), whereas Windows uses backslashes (\) (e.g., C:\Users\User\Documents).
- **Case Sensitivity:** Linux paths and file names are strictly case-sensitive (meaning File.txt and file.txt are treated as two entirely different files). Traditional Windows file systems (like NTFS) are generally case-insensitive by default.

2. Permissions (r, w, x)

Linux permissions define access levels for files and directories:

- **r (Read):** Allows viewing file contents or listing directory contents.
- **w (Write):** Allows modifying, deleting, or renaming files/directories.
- **x (Execute):** Allows running a file as a program or entering a directory.

Permissions apply to the **Owner** (file creator), **Group** (users in the same group), and **Others** (everyone else). Use `ls -l` to view these settings and `chmod` (e.g., `chmod 755 filename`) to modify them.

1. Read, Write, and Execute Permissions (r, w, x)

Linux permissions define the access levels granted to users for files and directories:

- **r (Read):** Allows viewing the contents of a file or listing the items inside a directory.
- **w (Write):** Allows modifying, deleting, or renaming files and directories.
- **x (Execute):** Allows running a file as an executable program or entering/accessing a directory.

2. Permission Categories (Owner, Group, and Others)

Permissions are assigned to three distinct classes of users:

- **Owner:** The user who created the file or directory, having primary control over it.
- **Group:** A set of users who share access rights to the file or directory, useful for collaboration.
- **Others:** All other users on the system who are neither the owner nor part of the authorized group.

3. Viewing and Changing Permissions

- **Viewing Permissions:** You can view the current permissions of files and directories by running the command `ls -l` in the terminal. This displays a detailed list showing permission bits for the owner, group, and others.
- **Changing Permissions:** You can modify or update permissions using the `chmod` command (for example, running `chmod 755 filename`).

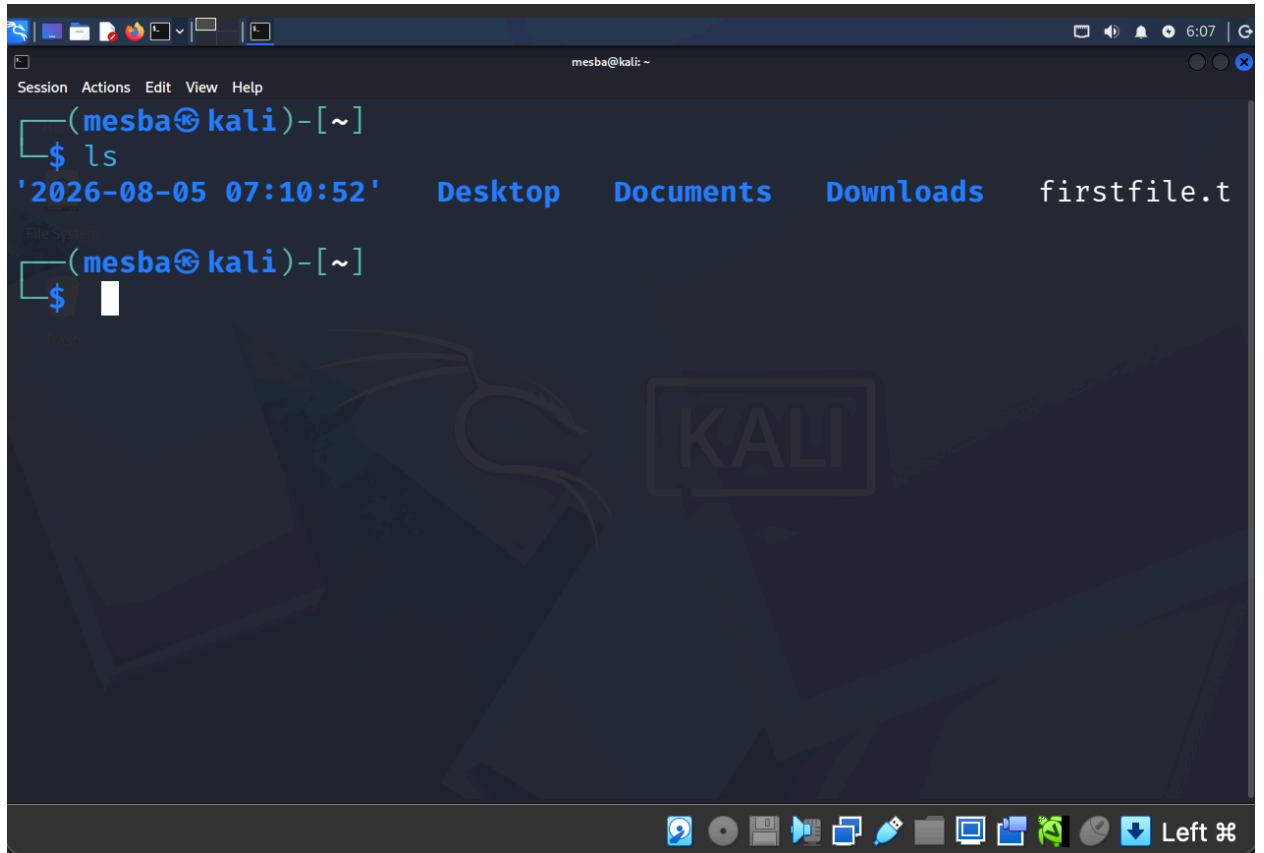
3. CLI vs. GUI

The Command Line Interface (CLI) is generally faster, requires fewer system resources, and allows for complex automation via scripts compared to the Graphical User Interface (GUI).

Why Cybersecurity Professionals prefer CLI: It provides granular control over system processes, enables quick parsing of data (using tools like grep), allows for remote management over SSH, and is essential for utilizing powerful terminal-based security tools efficiently.

1. `ls`

- **Description:** Lists directory contents.
- **Important Flags:** `-l` (long format), `-a` (show hidden)
- **Example:** `ls -la`



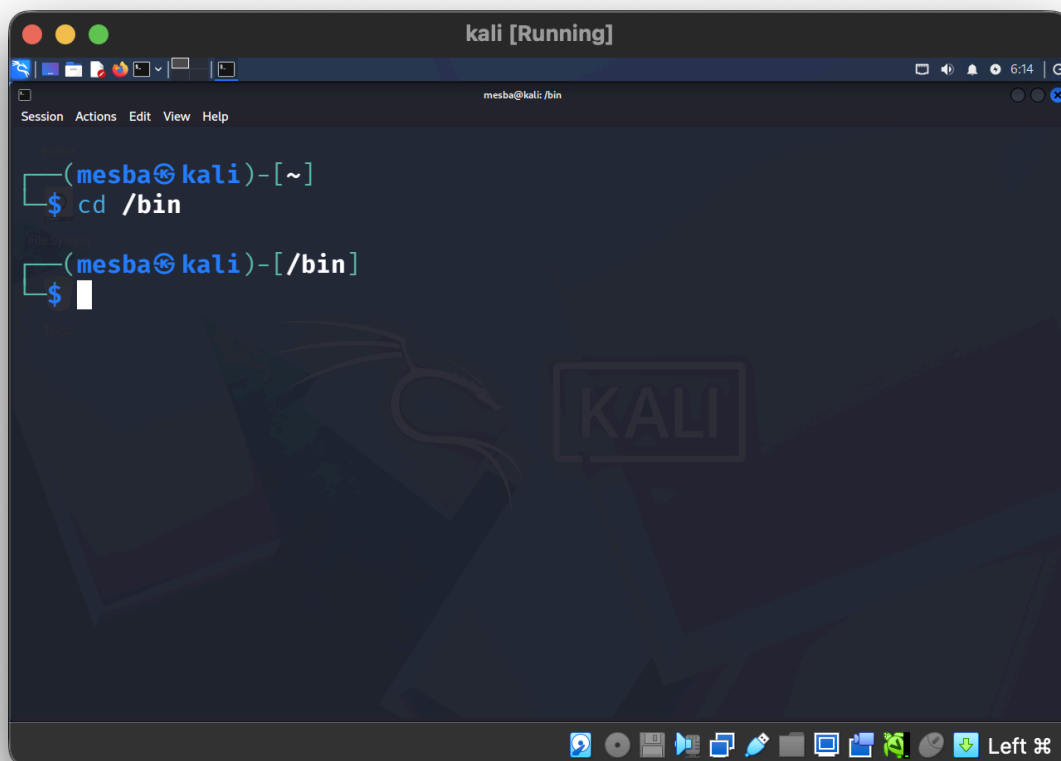
The screenshot shows a Kali Linux terminal window with a dark theme. The window title is 'mesba@kali: ~'. The terminal displays the following commands and output:

```
(mesba@kali)-[~]  
$ ls  
'2026-08-05 07:10:52' Desktop Documents Downloads firstfile.t  
(mesba@kali)-[~]  
$
```

The background of the terminal features a faint Kali Linux logo. The window's top bar includes standard Linux window controls and system status icons. The bottom bar shows various application icons and the text 'Left ⌘'.

2. `cd`

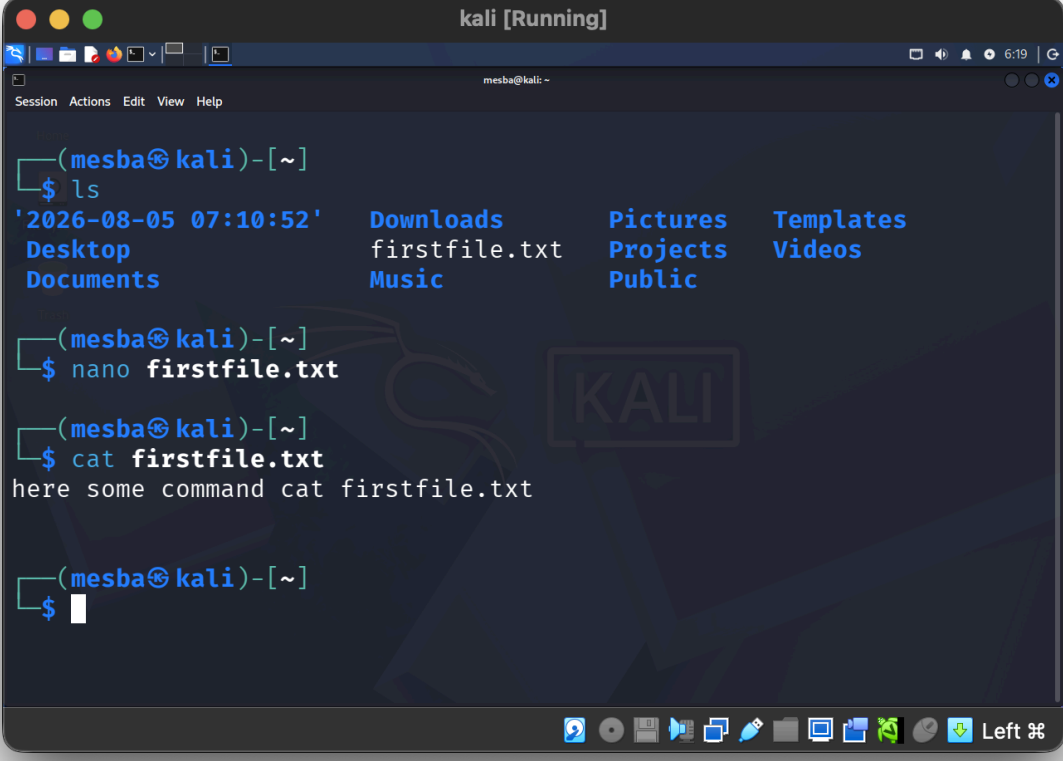
- **Description:** Change directory.
- **Important Flags:** `..` (parent dir), `~` (home)
- **Example:** `cd /var/log`



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3. `cat`

- **Description:** Concatenate and display file content.
- **Important Flags:** `-n` (number lines)
- **Example:** `cat file.txt`



The screenshot shows a Kali Linux terminal window titled 'kali [Running]'. The user 'mesba' is logged in. The terminal displays the following commands and output:

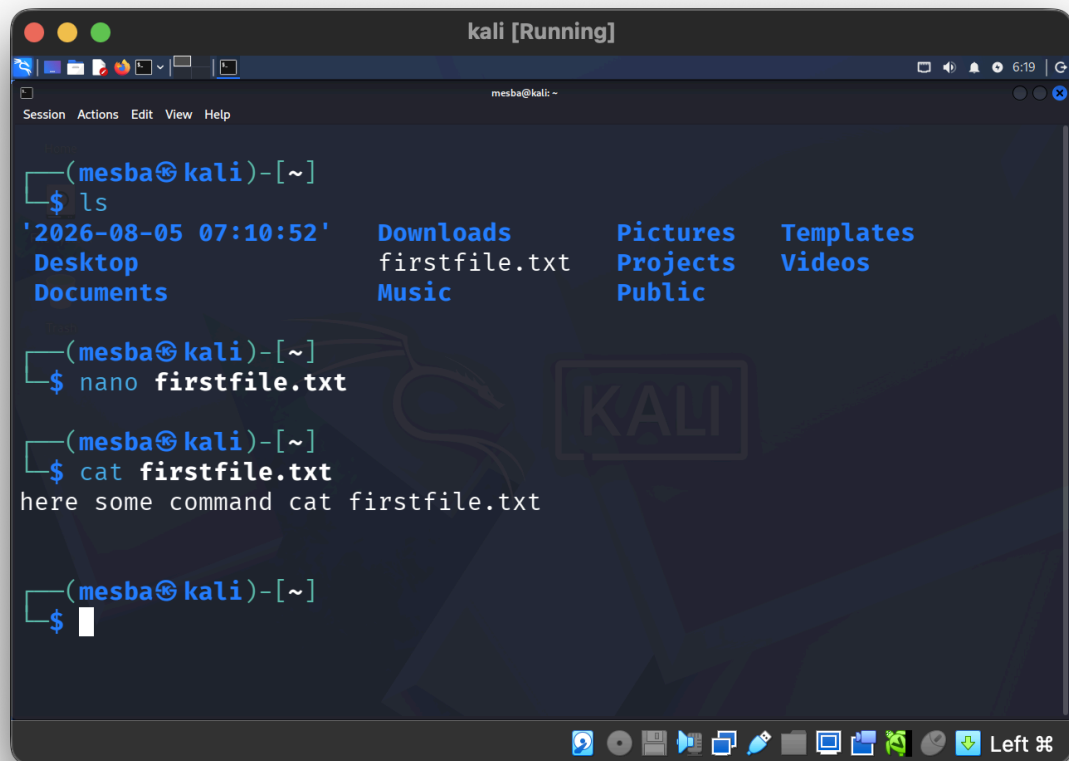
```
(mesba@kali)-[~]  
$ ls  
'2026-08-05 07:10:52' Downloads Pictures Templates  
Desktop firstfile.txt Projects Videos  
Documents Music Public  
  
(mesba@kali)-[~]  
$ nano firstfile.txt  
  
(mesba@kali)-[~]  
$ cat firstfile.txt  
here some command cat firstfile.txt  
  
(mesba@kali)-[~]  
$
```

The terminal background features a faint Kali Linux logo. The window has a standard macOS-style title bar with red, yellow, and green buttons. The bottom of the window shows a dock with various application icons and a system status bar with the text 'Left ⌘'.

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4. nano

- **Description:** A terminal-based text editor.
- **Important Flags:** N/A
- **Example:** nano file.txt



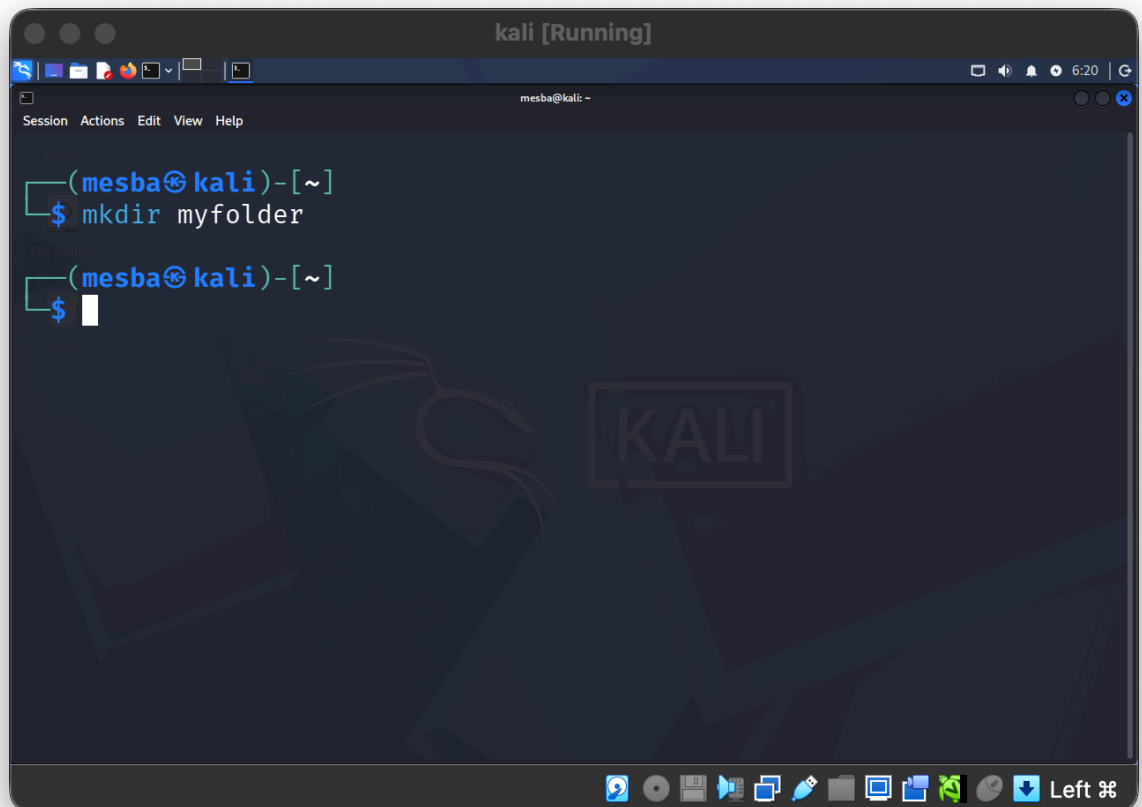
The screenshot shows a Kali Linux terminal window titled 'kali [Running]'. The user 'mesba' is logged in. The terminal displays the following commands and output:

```
(mesba@kali)-[~]  
$ ls  
'2026-08-05 07:10:52' Downloads Pictures Templates  
Desktop firstfile.txt Projects Videos  
Documents Music Public  
  
(mesba@kali)-[~]  
$ nano firstfile.txt  
  
(mesba@kali)-[~]  
$ cat firstfile.txt  
here some command cat firstfile.txt  
  
(mesba@kali)-[~]  
$
```

The terminal background features a faint Kali Linux logo. The window has a standard macOS-style title bar with red, yellow, and green buttons. The bottom of the window shows a dock with various application icons and a system status bar with the text 'Left ⌘'.

5. **mkdir**

- **Description:** Create a new directory.
- **Important Flags:** **-p** (create parent dirs)
- **Example:** **mkdir my_folder**



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6. **rm**

- **Description:** Remove files or directories.
- **Important Flags:** **-r** (recursive), **-f** (force)
- **Example:** **rm -rf folder_name**

The screenshot shows a Kali Linux terminal window titled "kali [Running]". The user "mesba" is logged in. The terminal shows the following commands and output:

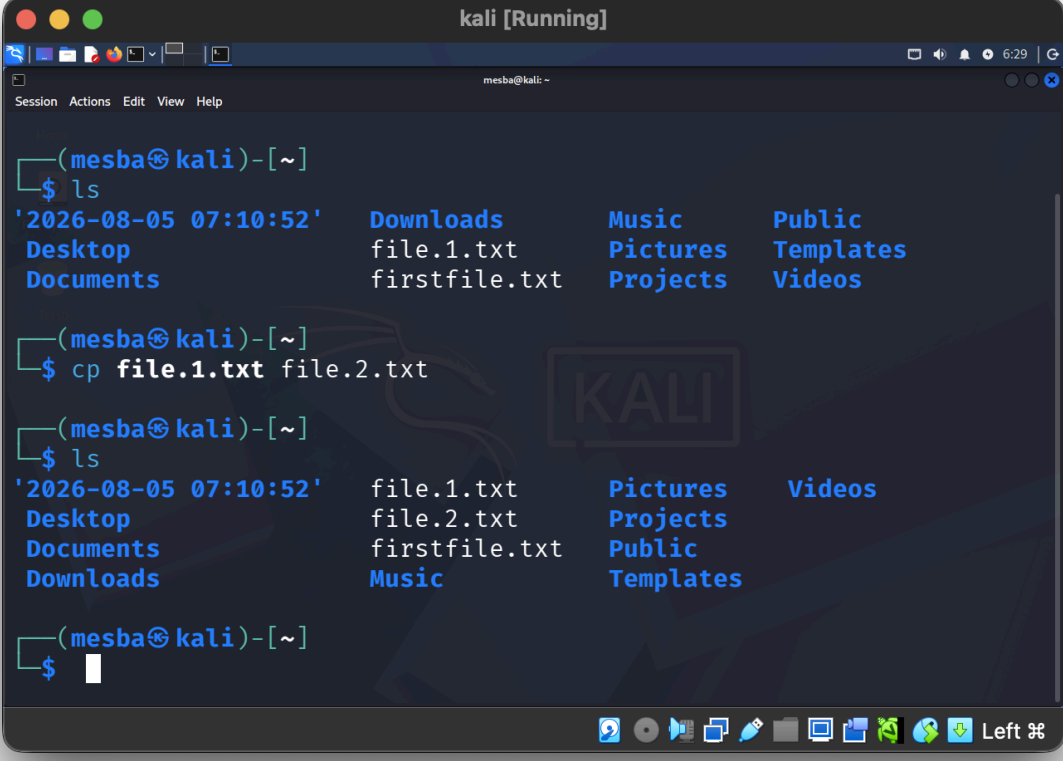
```
(mesba@kali)-[~]  
$ mkdir myfolder  
  
(mesba@kali)-[~]  
$ rm myfolder  
rm: cannot remove 'myfolder': Is a directory  
  
(mesba@kali)-[~]  
$ rm -rf myfolder  
  
(mesba@kali)-[~]  
$ ls  
'2026-08-05 07:10:52'  Downloads  Pictures  Templates  
Desktop               firstfile.txt Projects  Videos  
Documents             Music      Public
```

The terminal window has a menu bar with "Session", "Actions", "Edit", "View", and "Help". The status bar at the bottom shows various system icons and the text "Left ⌘".

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7. **cp**

- **Description:** Copy files and directories.
- **Important Flags:** **-r** (recursive)
- **Example:** **cp** file1 file2



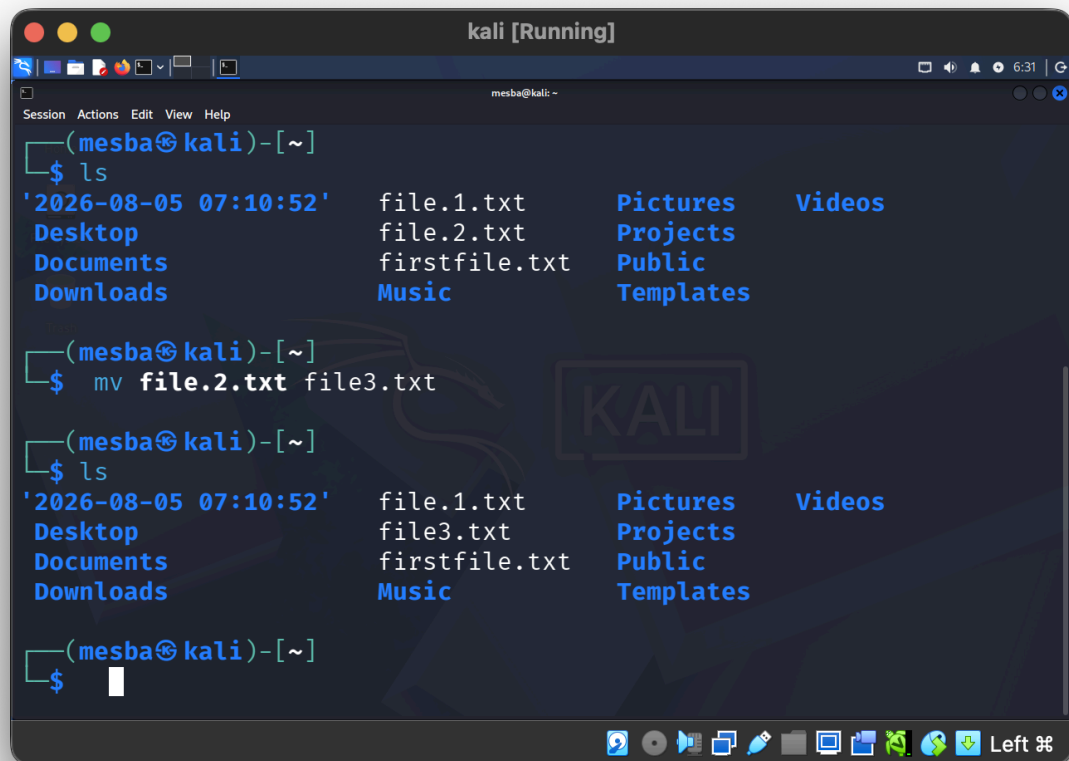
The screenshot shows a Kali Linux terminal window titled "kali [Running]". The user is logged in as "mesba" and is in the home directory (~). The terminal displays the following commands and output:

```
(mesba@kali)-[~]  
$ ls  
'2026-08-05 07:10:52' Downloads Music Public  
Desktop file.1.txt Pictures Templates  
Documents firstfile.txt Projects Videos  
  
(mesba@kali)-[~]  
$ cp file.1.txt file.2.txt  
  
(mesba@kali)-[~]  
$ ls  
'2026-08-05 07:10:52' file.1.txt Pictures Videos  
Desktop file.2.txt Projects  
Documents firstfile.txt Public  
Downloads Music Templates  
  
(mesba@kali)-[~]  
$
```

The terminal window has a menu bar with "Session", "Actions", "Edit", "View", and "Help". The status bar at the bottom shows various system icons and the text "Left ⌘".

8. mv

- **Description:** Move or rename files/directories.
- **Important Flags:** `-i` (interactive)
- **Example:** `mv old.txt new.txt`



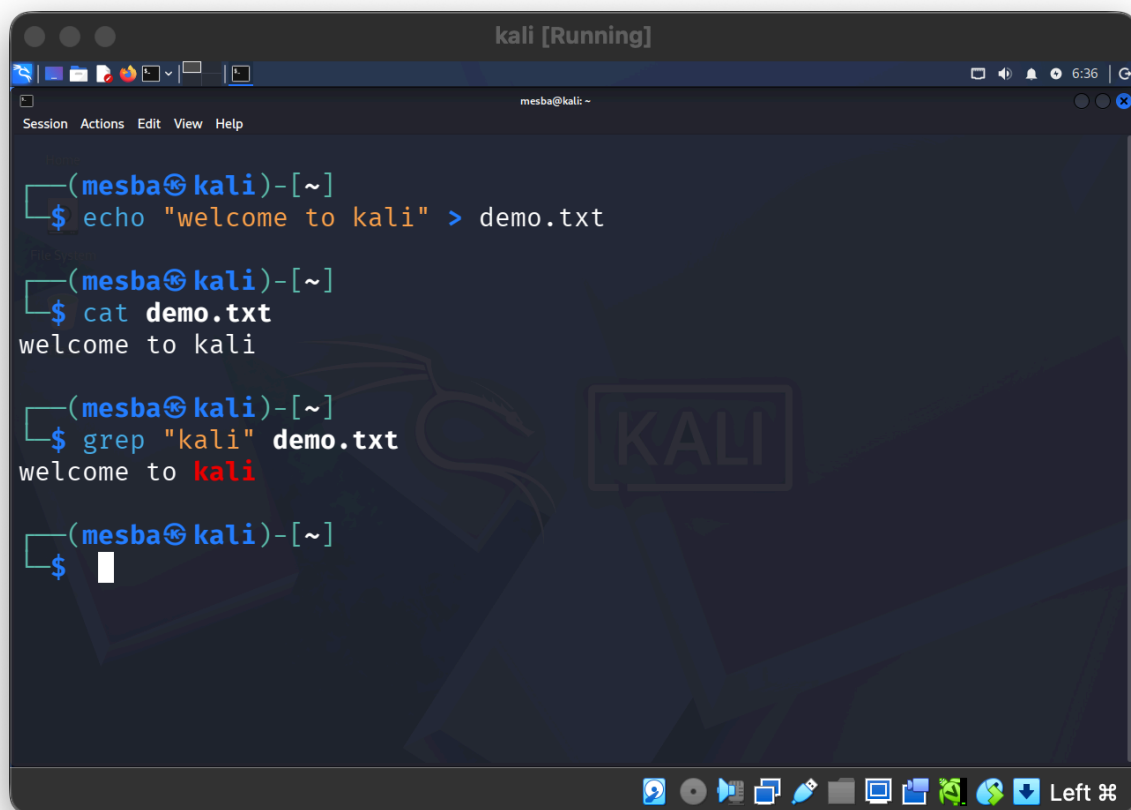
The screenshot shows a Kali Linux terminal window titled "kali [Running]". The user is logged in as "mesba" and is in the home directory (~). The terminal shows the following sequence of commands and outputs:

```
(mesba@kali)-[~]  
$ ls  
'2026-08-05 07:10:52' file.1.txt Pictures Videos  
Desktop file.2.txt Projects  
Documents firstfile.txt Public  
Downloads Music Templates  
  
(mesba@kali)-[~]  
$ mv file.2.txt file3.txt  
  
(mesba@kali)-[~]  
$ ls  
'2026-08-05 07:10:52' file.1.txt Pictures Videos  
Desktop file3.txt Projects  
Documents firstfile.txt Public  
Downloads Music Templates  
  
(mesba@kali)-[~]  
$
```

The terminal output shows a directory listing with files and directories. The files listed are file.1.txt, file.2.txt, firstfile.txt, and file3.txt. The directories listed are Desktop, Documents, Downloads, Music, Pictures, Projects, Public, Templates, and Videos. The user has successfully moved file.2.txt to file3.txt.

9. grep

- **Description:** Search text using patterns.
- **Important Flags:** `-i` (ignore case), `-r` (recursive)
- **Example:** `grep "search" file.txt`

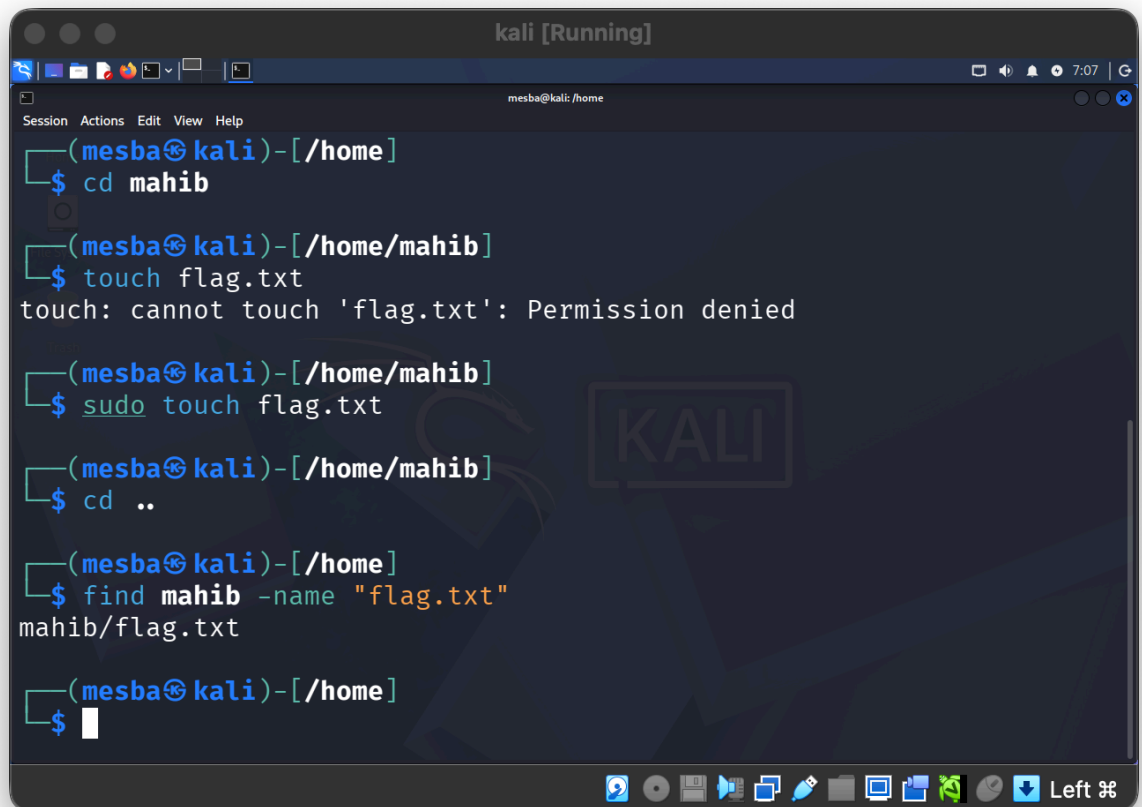


The image shows a terminal window titled "kali [Running]" with a menu bar (Session, Actions, Edit, View, Help) and a status bar (mesba@kali: ~, 6:36). The terminal displays the following commands and output:

```
(mesba@kali)-[~]  
$ echo "welcome to kali" > demo.txt  
  
(mesba@kali)-[~]  
$ cat demo.txt  
welcome to kali  
  
(mesba@kali)-[~]  
$ grep "kali" demo.txt  
welcome to kali  
  
(mesba@kali)-[~]  
$
```

10. find

- **Description:** Search for files in a directory hierarchy.
- **Important Flags:** `-name` (search by file name)
- **Example:** `find . -name "*.txt"`



```
kali [Running]
mesba@kali: /home

(mesba@kali)-[/home]
$ cd mahib

(mesba@kali)-[/home/mahib]
$ touch flag.txt
touch: cannot touch 'flag.txt': Permission denied

(mesba@kali)-[/home/mahib]
$ sudo touch flag.txt

(mesba@kali)-[/home/mahib]
$ cd ..

(mesba@kali)-[/home]
$ find mahib -name "flag.txt"
mahib/flag.txt

(mesba@kali)-[/home]
$
```